A Web Platform to Upload an Image and Construct Its 3D Model

Course Project	3D Computer Vision
Project type	Web application + 3D reconstruction pipeline (course project)
Core idea	User logs in, uploads an image (single or multi-view), and the system constructs a 3D model from it
Output formats	Point cloud, textured mesh, GLB/OBJ/PLY (previewable in browser)
Reference base	GitHub repos scraped & ranked by stars — direct public links

Reference document built by scraping GitHub for image-to-3D construction repositories. All GitHub links are public and directly accessible: https://github.com//.

1. Introduction and Problem Statement

Most 3D content today is created by hand in tools such as Blender, Maya, or CAD software, which requires specialized skills and significant time. This project builds an end-to-end web platform that lets any registered user upload an ordinary 2D photograph of an object or scene and automatically receive a reconstructed 3D model (point cloud, mesh, or Gaussian-splatted representation) that can be inspected in the browser and downloaded. The project is scoped as a 3D Computer Vision course project: it must demonstrate understanding of image formation, camera geometry, depth from images, multi-view geometry, neural 3D representations, and a working full-stack demo.

Problem statement: Given one or more 2D images of an object, estimate its 3D geometry and appearance through a web interface with user accounts, and present the result as an interactive, downloadable 3D model.

2. Project Objectives

- Build a responsive website with user registration, login, and protected project pages.
- Allow a logged-in user to upload one image (single-view mode) or several images of the same object (multi-view mode).
- Run a 3D reconstruction pipeline on the uploaded image(s) to produce geometry (depth map → point cloud → mesh, or neural radiance field / Gaussian splatting).
- Render the reconstructed model interactively in the browser (orbit, zoom, pan) using a WebGL viewer.
- Let the user download the model in standard formats (GLB, OBJ, PLY) and view a history of their reconstructions.
- Relate every technical choice to topics in a 3D Computer Vision syllabus (camera models, epipolar geometry, SfM/MVS, depth estimation, NeRF, Gaussian splatting, differentiable rendering).
- Document the pipeline, compare against open-source reference repositories, and evaluate reconstruction quality.

3. Scope and Features

In scope:

- Web frontend (React/Next.js or plain HTML+JS) with auth pages (signup/login/logout) and a dashboard.
- Image upload with validation, preview, and per-user job queue/history.
- Reconstruction backend (Python/FastAPI or Flask) calling open-source 3D-vision models.
- Depth estimation from a single image (e.g., MiDaS / Depth Anything) for single-view mode.
- Structure-from-Motion + Multi-View Stereo (COLMAP-style) for multi-view mode.
- Optional neural path: feed-forward single-image models (TripoSR, InstantMesh, SF3D, TRELLIS, Zero123) or per-scene optimization (Instant-NGP, 3D Gaussian Splatting).
- Browser-based 3D viewer (Three.js /) with orbit controls.
- Export to GLB/OBJ/PLY and per-user download history.

Out of scope (course project):

- Training new deep-learning models from scratch (use pretrained/open-weight models).
- Sub-millimetre metrological accuracy or drone/aerial photogrammetry at city scale.
- Paid third-party reconstruction APIs as the core method (may be used only as a fallback benchmark).

4. System Architecture

The platform is split into four logical layers. Each layer maps to a set of course topics:

Layer	Responsibility	Suggested technology	Course link
Client (website)	Login/signup, dashboard, image upload drag-and-drop, 3D viewer, download	React/Next.js or vanilla JS; Tailwind CSS; Three.js or ; Auth (email/password or OAuth)	Human-computer interaction; visualization
API / backend	Auth sessions, job queue, image storage, orchestration of reconstruction jobs, result metadata	FastAPI or Flask (Python); SQLite/PostgreSQL; AWS S3 or local disk; Celery/RQ or simple background jobs	Software engineering; databases
Reconstruction service	Preprocessing, depth estimation, SfM/MVS or neural reconstruction, mesh post-processing, GLB export	Python; OpenCV, COLMAP (optional), PyTorch; MiDaS/Depth Anything; TripoSR/InstantMesh/SF3D/TRELLIS; Instant-NGP or 3DGS (optional)	Camera geometry; depth; SfM; MVS; NeRF; Gaussian splatting
Viewer & export	Interactive WebGL rendering, orbit controls, GLB/OBJ/PLY download	Three.js, GLTFLoader, OrbitControls	Rendering; 3D file formats

5. End-to-End Technical Pipeline

Two supported modes share the same web front end but differ in the reconstruction engine:

Step	Single-view mode (one image)	Multi-view mode (several images)
1. Upload & auth	User logs in and uploads one photo	User logs in and uploads 5–30 photos around the object
2. Preprocessing	Resize, EXIF orientation, optional background removal	Rename, downscale if needed, EXIF parsing
3. Geometry	Monocular depth estimation (MiDaS / Depth Anything) → point cloud from camera intrinsics	Feature detection (SIFT/SIFT-like) → matching → SfM (camera poses) → MVS dense reconstruction
4. 3D representation	Depth map → point cloud → Poisson/textured mesh; optionally feed-forward model (TripoSR / SF3D / InstantMesh / TRELLIS) → Gaussian splat or mesh	Dense point cloud → meshing (Poisson) → textured mesh; optional per-object NeRF / 3D Gaussian Splatting refinement
5. Render & export	Three.js viewer; export GLB/OBJ/PLY	Same viewer; export textured GLB/OBJ/PLY
6. Save & history	Store result + source image in user's job history	Same

Representations produced: point cloud (.ply), textured mesh (.obj/.glb), optionally radiance field or 3D Gaussian Splatting checkpoint for novel-view rendering.

6. Suggested Technology Stack

Component	Suggested tools
Frontend	React / Next.js, Tailwind CSS; Three.js or for WebGL viewing
Authentication	Email/password sessions or OAuth (NextAuth / Firebase / Flask-Login)
Backend API	FastAPI or Flask (Python) with job queue (Celery/RQ or simple background worker)
Database	PostgreSQL or SQLite for users, jobs, and result metadata
Storage	AWS S3 / MinIO (or local disk) for images and exported 3D files
Depth (single view)	MiDaS, Depth Anything V2, or DPT via Hugging Face (PyTorch)
SfM / MVS (multi view)	COLMAP or OpenCV-based feature matching + triangulation (optional)
Feed-forward 3D models	TripoSR, Stable Fast 3D (SF3D), InstantMesh, TRELLIS, Zero123 / Zero123++
Per-scene optimization (optional)	Instant-NGP or 3D Gaussian Splatting
Mesh post-processing / export	Open3D, Trimesh, GLTF/GLB exporter
Viewer & export	Three.js, OrbitControls; export GLB/OBJ/PLY

7. Algorithms and Models to Investigate

The reconstruction engine can be chosen per mode. The table below groups the main algorithm families:

Family	What it does	Example open-source repos
Monocular depth	Predicts a depth map from one image; back-project to a point cloud using camera intrinsics	MiDaS, Depth Anything V2, DPT
SfM / MVS (multi-view)	Estimate camera poses from feature matches, then dense 3D from multiple views	COLMAP, OpenCV SfM, Meshroom
Feed-forward single-image 3D	One forward pass from image to 3D (mesh, radiance field, or Gaussian splat)	TripoSR, Stable Fast 3D, InstantMesh, TRELLIS, Zero123 / Zero123++
Per-scene neural rendering	Optimize a scene representation from images; novel-view synthesis	Instant-NGP, 3D Gaussian Splatting, Stable Diffusion DreamFusion
Classic single-view	Early learning-based reconstruction to voxels from one/multiple views	3D-R2N2, Pix2Vox

8. Reference GitHub Repositories — Scraped & Ranked by Stars

The repositories below were scraped from GitHub's public search API using image-to-3D construction queries, filtered for relevance, and ranked by stars. All links are public and directly accessible. Repos with a website / web UI / viewer component are especially relevant to your login + upload + 3D viewer goal.

8a. Web-App / Website Repos (login + upload image + 3D viewer) — closest to your goal

These repositories already combine a website/web interface with image-to-3D construction. They are the best architectural references.

Stars	Repository	Owner	Description	Direct GitHub Link
7300	**TripoSR**	VAST-AI-Research	TripoSR: Fast 3D Object Reconstruction from a Single Image	https://github.com/VAST-AI-Research/TripoSR
1087	**stable-point-aware-3d**	Stability-AI	SPAR3D: Stable Point-Aware Reconstruction of 3D Objects from Single Images	https://github.com/Stability-AI/stable-point-aware-3d
768	**3DGenStudio**	visualbruno	Orchestrate complete 3D generation pipelines — from text-to-image, image editing, mesh generation, UV unwrapping, to tex	https://github.com/visualbruno/3DGenStudio
724	**AtlasNet**	ThibaultGROUEIX	This repository contains the source codes for the paper "AtlasNet: A Papier-Mâché Approach to Learning 3D Surface Genera	https://github.com/ThibaultGROUEIX/AtlasNet
659	**3d-book-image-css-generator**	scastiel	Generate a 3D image from a book cover and export to HTML/CSS to embed on your website.	https://github.com/scastiel/3d-book-image-css-generator
570	**PointSetGeneration**	fanhqme	Code for ``A Point Set Generation Network for 3D Object Reconstruction from a Single Image''	https://github.com/fanhqme/PointSetGeneration
543	**Pix2Vox**	hzxie	The official implementation of "Pix2Vox: Context-aware 3D Reconstruction from Single and Multi-view Images". (ICCV 2019)	https://github.com/hzxie/Pix2Vox
491	**TF_FLAME**	TimoBolkart	Tensorflow framework for the FLAME 3D head model. The code demonstrates how to sample 3D heads from the model, fit the m	https://github.com/TimoBolkart/TF_FLAME
446	**Total3DUnderstanding**	GAP-LAB-CUHK-SZ	Implementation of CVPR'20 Oral: Total3DUnderstanding: Joint Layout, Object Pose and Mesh Reconstruction for Indoor Scene	https://github.com/GAP-LAB-CUHK-SZ/Total3DUnderstanding
376	**trellis.cpp**	pwilkin	TRELLIS.2 image-to-3D in C++/GGML (CUDA + Vulkan), with a resident HTTP server	https://github.com/pwilkin/trellis.cpp
308	**TouchTerrain_for_CAGEO**	ChHarding	Touch Terrain: A python app to create 3D printable terrain models (STL/OBJ) from only elevation data (via Google Earth E	https://github.com/ChHarding/TouchTerrain_for_CAGEO

Stars	Repository	Owner	Description	Direct GitHub Link
249	**3DObjectReconstruction**	NVIDIA	3D Object Reconstruction project is a workflow that takes a set of stereo images and camera info and outputs a textured	https://github.com/NVIDIA/3DObjectReconstruction
179	**CVPR2018_attention**	USTCPCS	Context Encoding for Semantic Segmentation MegaDepth: Learning Single-View Depth Prediction from Internet Photos LiteF	https://github.com/USTCPCS/CVPR2018_attention
145	**MVDiffusion_plusplus**	Tangshitao	MVDiffusion++: A Dense High-resolution Multi-view Diffusion Model for Single or Sparse-view 3D Object Reconstruction	https://github.com/Tangshitao/MVDiffusion_plusplus
144	**Multiview-3D-Reconstruction**	Yashas120	Using traditional image processing techniques to construct 3D point cloud of objects. Incremental Structure from Motion	https://github.com/Yashas120/Multiview-3D-Reconstruction
106	**guidance-for-open-source-3d-reconstruction-toolbox-for-gaussian-splats-on-aws**	aws-solutions-library-samples	The Open Source 3D Reconstruction Toolbox for Gaussian Splats provides an end-to-end, pipeline-based guidance on AWS to	https://github.com/aws-solutions-library-samples/guidance-for-open-source-3d-reconstruction-toolbox-for-gaussian-splats-on-aws
105	**pytorch3d-renderer**	adele-k02	A PyTorch3D walkthrough and a Medium article ■ on how to render 3D .obj meshes from various viewpoints to create 2D imag	https://github.com/adele-k02/pytorch3d-renderer
97	**CenterPoint-Fusion**	yinjunbo	The proposed approach enhances the CenterPoint baseline with a multimodal fusion mechanism. First, inspired by PointPain	https://github.com/yinjunbo/CenterPoint-Fusion
92	**3D-Object-Reconstrution**	wkqscut	A curated list of awesome Single-view 3D Object Reconstruction papers & resources	https://github.com/wkqscut/3D-Object-Reconstrution
77	**trellis_blender**	FishWoWater	Blender plugin for TRELLIS and TRELLIS.2 (3D AIGC Model, Text-to-3D, Image-to-3D)	https://github.com/FishWoWater/trellis_blender
72	**SharpGlass**	TrondW	SharpGlass is a macOS application for generating and rendering 3D Gaussian Splats from single images using Apple's ml-sh	https://github.com/TrondW/SharpGlass
67	**showcase-images**	cortiz2894	An immersive 3D scene built with Three.js, designed to showcase images in a dynamic and interactive way. The gallery rea	https://github.com/cortiz2894/showcase-images
67	**obj2png**	pclausen	Python script for converting 3D triangular mesh (obj) to image file (png) with MatPlotLib	https://github.com/pclausen/obj2png
65	**sphr**	lukehollis	Open, hackable 3d digital twin builder that supports 3d gaussian splatting, 360 images, and vfx	https://github.com/lukehollis/sphr
62	**TripoSR-webui**	Z-L-D	A TriposR implementation for WebUI	https://github.com/Z-L-D/TripoSR-webui

8b. All Scraped Repos — Ranked by Stars (full list)

Total repositories scraped and ranked: **343**. Sorted by GitHub stars (descending).

#	Stars	Repository	Owner	Description	Direct GitHub Link
1	8876	**stable-dreamfusion**	ashawkey	Text-to-3D & Image-to-3D & Mesh Exportation with NeRF + Diffusion.	https://github.com/ashawkey/stable-dreamfusion
2	7300	**TripoSR**	VAST-AI-Research	TripoSR: Fast 3D Object Reconstruction from a Single Image	https://github.com/VAST-AI-Research/TripoSR
3	6586	**ODM**	OpenDroneMap	A command line toolkit to generate maps, point clouds, 3D models and DEMs from drone, balloon o	https://github.com/OpenDroneMap/ODM
4	4591	**InstantMesh**	TencentARC	InstantMesh: Efficient 3D Mesh Generation from a Single Image with Sparse-view Large Reconstruc	https://github.com/TencentARC/InstantMesh
5	4523	**vrn**	AaronJackson	:man: Code for "Large Pose 3D Face Reconstruction from a Single Image via Direct Volumetric CN	https://github.com/AaronJackson/vrn

#	Stars	Repository	Owner	Description	Direct GitHub Link
6	4438	**awesome_3DReconstruction_list**	openMVG	A curated list of papers & resources linked to 3D reconstruction from images.	https://github.com/openMVG/awesome_3DReconstruction_list
7	2356	**Deep3DFaceReconstruction**	microsoft	Accurate 3D Face Reconstruction with Weakly-Supervised Learning: From Single Image to Image Set	https://github.com/microsoft/Deep3DFaceReconstruction
8	2246	**ooosplat**	ooolabdev	A local desktop app that turns videos and images into 3D Gaussian Splats in one click.	https://github.com/ooolabdev/ooosplat
9	1913	**Deep3DFaceRecon_pytorch**	sicxu	Accurate 3D Face Reconstruction with Weakly-Supervised Learning: From Single Image to Image Set	https://github.com/sicxu/Deep3DFaceRecon_pytorch
10	1860	**Atlas**	magicleap	Atlas: End-to-End 3D Scene Reconstruction from Posed Images	https://github.com/magicleap/Atlas
11	1766	**Pixel2Mesh**	nywang16	Pixel2Mesh: Generating 3D Mesh Models from Single RGB Images. In ECCV2018.	https://github.com/nywang16/Pixel2Mesh
12	1715	**One-2-3-45**	One-2-3-45	[NeurIPS 2023] Official code of "One-2-3-45: Any Single Image to 3D Mesh in 45 Seconds without	https://github.com/One-2-3-45/One-2-3-45
13	1410	**3D-R2N2**	chrischoy	Single/multi view image(s) to voxel reconstruction using a recurrent neural network	https://github.com/chrischoy/3D-R2N2
14	1322	**mvsplat**	donydchen	■ [ECCV'24 Oral] MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images	https://github.com/donydchen/mvsplat
15	1285	**pixelsplat**	dcharatan	[CVPR 2024 Oral, Best Paper Runner-Up] Code for "pixelSplat: 3D Gaussian Splats from Image Pair	https://github.com/dcharatan/pixelsplat
16	1116	**splatter-image**	szymanowiczs	Official implementation of `Splatter Image: Ultra-Fast Single-View 3D Reconstruction' CVPR 2024	https://github.com/szymanowiczs/splatter-image
17	1112	**mvs-texturing**	nmoehrle	Algorithm to texture 3D reconstructions from multi-view stereo images	https://github.com/nmoehrle/mvs-texturing
18	1107	**AdelaiDepth**	aim-uofa	This repo contains the projects: 'Virtual Normal', 'DiverseDepth', and '3D Scene Shape'. They a	https://github.com/aim-uofa/AdelaiDepth
19	1095	**text2room**	lukasHoel	Text2Room generates textured 3D meshes from a given text prompt using 2D text-to-image models (	https://github.com/lukasHoel/text2room
20	1087	**stable-point-aware-3d**	Stability-AI	SPAR3D: Stable Point-Aware Reconstruction of 3D Objects from Single Images	https://github.com/Stability-AI/stable-point-aware-3d
21	1006	**NoPoSplat**	cvg	[ICLR'25 Oral] No Pose, No Problem: Surprisingly Simple 3D Gaussian Splats from Sparse Unposed	https://github.com/cvg/NoPoSplat
22	927	**TriplaneGaussian**	VAST-AI-Research	TriplaneGaussian: A new hybrid representation for single-view 3D reconstruction.	https://github.com/VAST-AI-Research/TriplaneGaussian
23	869	**Open-DiffusionGS**	caiyuanhao1998	Baking Gaussian Splatting into Diffusion Denoiser for Fast and Scalable Single-stage Image-to-3	https://github.com/caiyuanhao1998/Open-DiffusionGS
24	832	**panic3d-anime-reconstruction**	ShuhongChen	CVPR 2023: PAniC-3D Stylized Single-view 3D Reconstruction from Portraits of Anime Characters	https://github.com/ShuhongChen/panic3d-anime-reconstruction
25	768	**3DGenStudio**	visualbruno	Orchestrate complete 3D generation pipelines — from text-to-image, image editing, mesh generati	https://github.com/visualbruno/3DGenStudio
26	736	**I2L-MeshNet_RELEASE**	mks0601	Official PyTorch implementation of "I2L-MeshNet: Image-to-Lixel Prediction Network for Accurate	https://github.com/mks0601/I2L-MeshNet_RELEASE
27	724	**AtlasNet**	ThibaultGROUEIX	This repository contains the source codes for the paper "AtlasNet: A Papier-Mâché Approach to L	https://github.com/ThibaultGROUEIX/AtlasNet

#	Stars	Repository	Owner	Description	Direct GitHub Link
28	713	**LHM-plusplus**	aigc3d	LHM++: An Efficient Large Human Reconstruction Model for Pose-free Images to 3D	https://github.com/aigc3d/LHM-plusplus
29	704	**CRM**	thu-ml	[ECCV 2024] Single Image to 3D Textured Mesh in 10 seconds with Convolutional Reconstruction Mo	https://github.com/thu-ml/CRM
30	659	**3d-book-image-css-generator**	scastiel	Generate a 3D image from a book cover and export to HTML/CSS to embed on your website.	https://github.com/scastiel/3d-book-image-css-generator
31	601	**colmap-pcd**	XiaoBaiiiiii	Image-to-point cloud Registration Tool	https://github.com/XiaoBaiiiiii/colmap-pcd
32	570	**PointSetGeneration**	fanhqme	Code for ``A Point Set Generation Network for 3D Object Reconstruction from a Single Image''	https://github.com/fanhqme/PointSetGeneration
33	570	**FaceLift**	weijielyu	[ICCV 2025] FaceLift: Learning Generalizable Single Image 3D Face Reconstruction from Synthetic	https://github.com/weijielyu/FaceLift
34	553	**DiffSplat**	chenguolin	[ICLR 2025] Official implementation of "DiffSplat: Repurposing Image Diffusion Models for Scala	https://github.com/chenguolin/DiffSplat
35	543	**Pix2Vox**	hzxie	The official implementation of "Pix2Vox: Context-aware 3D Reconstruction from Single and Multi-	https://github.com/hzxie/Pix2Vox
36	536	**pixel3dmm**	SimonGiebenhain	[Official Code] Pixel3DMM: Versatile Screen-Space Priors for Single-Image 3D Face Reconstructio	https://github.com/SimonGiebenhain/pixel3dmm
37	491	**TF_FLAME**	TimoBolkart	Tensorflow framework for the FLAME 3D head model. The code demonstrates how to sample 3D heads	https://github.com/TimoBolkart/TF_FLAME
38	469	**2DASL**	XgTu	The code (pytorch for testing & matlab for 3D plot and evaluation) for our project: Joint 3D Fa	https://github.com/XgTu/2DASL
39	446	**Total3DUnderstanding**	GAP-LAB-CUHK-SZ	Implementation of CVPR'20 Oral: Total3DUnderstanding: Joint Layout, Object Pose and Mesh Recons	https://github.com/GAP-LAB-CUHK-SZ/Total3DUnderstanding
40	435	**Lift4D**	yehonathanlitman	[SIGGRAPH Asia 2026] Lift4D: Harmonizing Single-View 3D Estimation for 4D Reconstruction In-the	https://github.com/yehonathanlitman/Lift4D
41	429	**DeepHuman**	ZhengZerong	Code for our ICCV paper "DeepHuman: 3D Human Reconstruction from a Single Image"	https://github.com/ZhengZerong/DeepHuman
42	412	**Deblur-GS**	Chaphlagical	[I3D 2024] Deblur-GS: 3D Gaussian Splatting from Camera Motion Blurred Images	https://github.com/Chaphlagical/Deblur-GS
43	392	**pytorch-3d-point-cloud-generation**	lkhphuc	Pytorch code to construct a 3D point cloud model from single RGB image.	https://github.com/lkhphuc/pytorch-3d-point-cloud-generation
44	380	**vrn-docker**	AaronJackson	:whale: Docker files for Large Pose 3D Face Reconstruction from a Single Image via Direct Volum	https://github.com/AaronJackson/vrn-docker
45	376	**trellis.cpp**	pwilkin	TRELLIS.2 image-to-3D in C++/GGML (CUDA + Vulkan), with a resident HTTP server	https://github.com/pwilkin/trellis.cpp
46	376	**IDOL**	yiyuzhuang	IDOL: Instant Photorealistic 3D Human Creation from a Single Image. An open-source project for	https://github.com/yiyuzhuang/IDOL
47	372	**PlanarReconstruction**	svip-lab	[CVPR'19] Single-Image Piece-wise Planar 3D Reconstruction via Associative Embedding	https://github.com/svip-lab/PlanarReconstruction
48	337	**flash3d**	eldar	Flash3D: Feed-Forward Generalisable 3D Scene Reconstruction from a Single Image	https://github.com/eldar/flash3d
49	311	**RasterFairy**	Quasimondo	The purpose of Raster Fairy is to transform any kind of 2D point cloud into a regular raster wh	https://github.com/Quasimondo/RasterFairy
50	310	**FreeReg**	WHU-USI3DV	[ICLR 2024] FreeReg: Image-to-Point Cloud Registration Leveraging Pretrained Diffusion Models a	https://github.com/WHU-USI3DV/FreeReg

#	Stars	Repository	Owner	Description	Direct GitHub Link
51	310	**DISN**	laughtervv	DISN: Deep Implicit Surface Network for High-quality Single-view 3D Reconstruction	https://github.com/laughtervv/DISN
52	308	**TouchTerrain_for_CAGEO**	ChHarding	Touch Terrain: A python app to create 3D printable terrain models (STL/OBJ) from only elevation	https://github.com/ChHarding/TouchTerrain_for_CAGEO
53	308	**SingleViewReconstruction**	DLR-RM	Official Code: 3D Scene Reconstruction from a Single Viewport	https://github.com/DLR-RM/SingleViewReconstruction
54	275	**DepthRegularizedGS**	robot0321	This is an official repository for "Depth-Regularized Optimization for 3D Gaussian Splatting in	https://github.com/robot0321/DepthRegularizedGS
55	271	**OcclusionFusion**	wenbin-lin	OcclusionFusion: realtime dynamic 3D reconstruction based on single-view RGB-D	https://github.com/wenbin-lin/OcclusionFusion
56	269	**mesh_reconstruction**	hiroharu-kato	Code of single-image 3D reconstruction "Neural 3D Mesh Renderer" by H. Kato, Y. Ushiku, and T.	https://github.com/hiroharu-kato/mesh_reconstruction
57	261	**DeepI2P**	lijx10	DeepI2P: Image-to-Point Cloud Registration via Deep Classification. CVPR 2021	https://github.com/lijx10/DeepI2P
58	249	**3DObjectReconstruction**	NVIDIA	3D Object Reconstruction project is a workflow that takes a set of stereo images and camera inf	https://github.com/NVIDIA/3DObjectReconstruction
59	245	**Code-as-Room**	YxuanAr	A MLLM-based agentic system converts a single room image into executable Blender code for 3D ro	https://github.com/YxuanAr/Code-as-Room
60	234	**UMR**	NVlabs	Self-supervised Single-view 3D Reconstruction	https://github.com/NVlabs/UMR

Star counts reflect GitHub figures at scrape time (late 2025 / 2026) and are used only for ranking. Verify live counts on GitHub before citing. Full scraped dataset saved in data/repos_by_stars.json.

9. Suggested Milestones and Timeline (map to a 3D Computer Vision syllabus)

A typical course project runs over a semester. The milestones below are aligned with standard 3D computer vision topics so each deliverable reinforces syllabus content.

Week	Milestone	Syllabus topic covered	Deliverable
1–2	Requirement analysis, team setup, choose stack; scaffold website with signup/login/login-protected routes.	Course intro; project planning	Repo + live auth pages
3–4	Image upload + storage; dashboard showing user's jobs; basic backend job queue.	Software engineering; image formation	Upload flow + dashboard
5–6	Single-view path: run monocular depth (MiDaS/Depth Anything) → point cloud; render in Three.js.	Camera models; depth from single image	Depth map + point-cloud viewer
7–8	Multi-view path: feature matching + SfM camera poses → sparse point cloud (COLMAP/OpenCV).	Epipolar geometry; SfM	Multi-view pose + sparse cloud
9–10	Dense MVS + meshing → textured mesh; export GLB/OBJ/PLY; download.	Multi-view stereo; meshing	Textured mesh + export
11–12	Optional neural upgrade: feed-forward single-image model (TripoSR/SF3D/InstantMesh/TRELLIS) as an alternative engine; compare quality vs. depth/SfM path.	Neural 3D representations; NeRF/Gaussian splatting	Comparison report + second engine
13	Final polish: viewer improvements, history, documentation, final demo and report.	Rendering; presentation	Final website + report + demo video

10. Evaluation and Success Criteria

- Functional: a registered user can upload an image and receive a viewable, downloadable 3D model.
- Quality: visual fidelity of reconstruction vs. input (qualitative); for multi-view, reprojection error and point-cloud density; for single-view, depth consistency and mesh completeness.
- Performance: end-to-end job time for single-view (target seconds) vs. multi-view (target minutes).
- Course alignment: report must connect each module to 3D computer vision concepts (camera geometry, depth, SfM/MVS, neural rendering).
- References: at least 5–8 GitHub repositories from Sections 8a/8b used and cited with links.

11. Challenges and Risk Mitigation

Risk	Mitigation
Single-view depth is scale-ambiguous and hallucinates occluded parts.	State assumptions clearly; use multi-view mode for metric accuracy; label single-view output as "approx. shape".
Large models (TRELLIS, Instant-NGP) need a strong GPU.	Use CPU-friendly depth/SfM path as default; run heavy models on Google Colab / a GPU instance as an optional tier.
WebGL viewer fails on large meshes.	Simplify/decimate meshes before export; provide .ply/.obj fallback download.
User uploads poor images (blur, textureless, reflective objects).	Add image-quality tips in the UI and graceful error messages; require texture/overlap guidance for multi-view.
Authentication and job persistence add scope creep.	Use a ready auth solution (Firebase/NextAuth/Flask-Login) and SQLite to keep it minimal.

Next steps: share your course syllabus so the milestones and technical requirements can be tuned exactly to it, then pick one single-view engine (e.g., TripoSR or Depth Anything → Open3D) and one multi-view engine (COLMAP or OpenCV SfM) to start prototyping the website.

All GitHub links in Sections 8a and 8b are public and can be opened directly: https://github.com//.

Scraping program: github_scraper.py | Scraped data: data/repos_ranked.json, data/repos_by_stars.json